

Human analysis of Evidence03.pcap

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DAVE SUSHAMES

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Executive summary

The evidence03.pcap file contains approximately three minutes of network traffic. On 28 December 2009, between 04:08 and 04:10 UTC, an Apple TV (IP: 192.168.1.10) was used to browse the iTunes movie store. Three searches were made: One search for “hack” which led to the movie *Hackers*. One search for “sneak” which led to the movie *Sneakers*. One search for “iknowyourewatchingme”, which suggests that the user believed their network activity was being monitored.

Tools used

Wireshark version 4.6.9
md5sum (for hashing)
Greenshot (for screenshots)

Evidence description

File name	evidence03.pcap
MD5 hash	f8a01fbe84ef960d7cbd793e0c52a6c9
File type	Pcap
File size	1,537,222 bytes
Packets	1,778
First packet	Dec 28, 2009 04:08:01.139183000 UTC
Last packet	Dec 28, 2009 04:10:52.310744000 UTC
Duration	~171 seconds

Before analysis, the evidence file was hashed using MD5 Hashing the file again post-analysis returns the same result, proving that it wasn't edited in any way during analysis.

Device details

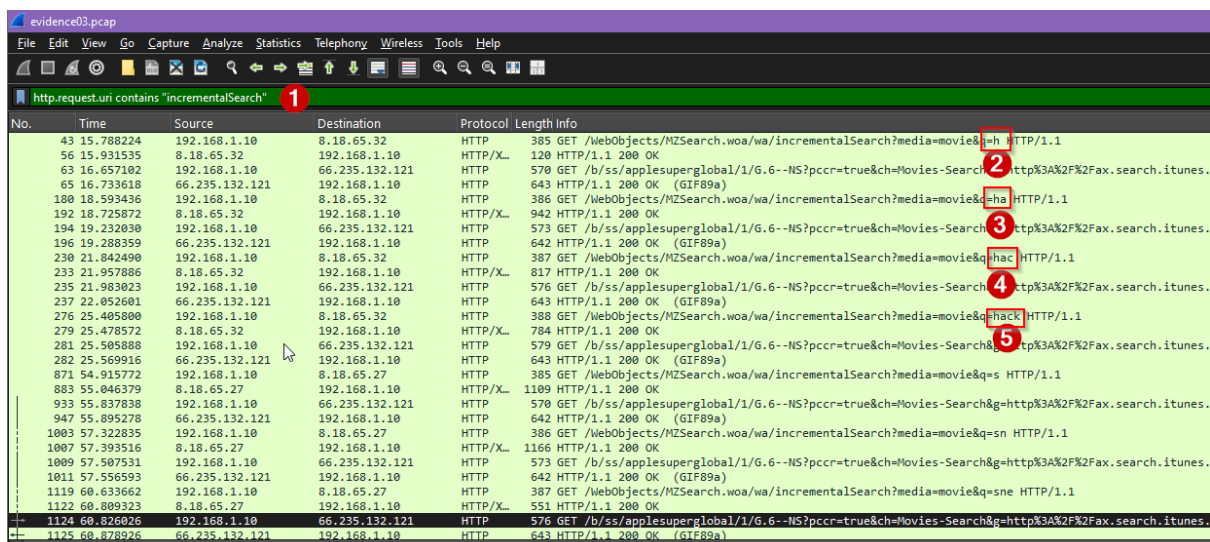
MAC address	00:25:00:fe:07:c4
IP address	192.168.1.10
Device type	Apple TV
Software	User-Agent: AppleTV/2.4
DNS server	4.2.2.1

The only internal IP listed in the capture was 192.168.1.10.

Findings

Figure 1 is a screenshot of the evidence file open in Wireshark, with the filter “http.request.uri contains “incrementalSearch”” applied (1). Displayed in frames 43, 180, 230, and 276 are searches for “h” (2), “ha” (3), “hac” (4), and “hack” (5) being typed in one letter at a time, over the span of 10 seconds.

Figure 1 - Hack search



The screenshot shows the Wireshark interface with the filter 'http.request.uri contains "incrementalSearch"' applied. The packet list shows several HTTP GET requests to 'http://www.apple.com/itunes/1.0/objects/MZSearch.wa/wa/incrementalSearch?media=movie&q=h' and similar queries with 'ha', 'hac', and 'hack'. The packet details pane shows the 'h' character being typed in the search query.

No.	Time	Source	Destination	Protocol	Length	Info
43	15.788224	192.168.1.10	8.18.65.32	HTTP	385	GET /WebObjects/MZSearch.wa/wa/incrementalSearch?media=movie&q=h HTTP/1.1
56	15.931535	8.18.65.32	192.168.1.10	HTTP/X.	120	HTTP/1.1 200 OK
63	16.657182	192.168.1.10	66.235.132.121	HTTP	570	GET /b/ss/applesuperglobal/1/G.6--NS?pcr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.
65	16.733618	66.235.132.121	192.168.1.10	HTTP	643	HTTP/1.1 200 OK (GIF89a)
180	18.593436	192.168.1.10	8.18.65.32	HTTP	386	GET /WebObjects/MZSearch.wa/wa/incrementalSearch?media=movie&q=ha HTTP/1.1
192	18.725872	8.18.65.32	192.168.1.10	HTTP/X.	942	HTTP/1.1 200 OK
194	19.232030	192.168.1.10	66.235.132.121	HTTP	573	GET /b/ss/applesuperglobal/1/G.6--NS?pcr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.
196	19.288359	66.235.132.121	192.168.1.10	HTTP	642	HTTP/1.1 200 OK (GIF89a)
230	21.842490	192.168.1.10	8.18.65.32	HTTP	387	GET /WebObjects/MZSearch.wa/wa/incrementalSearch?media=movie&q=hac HTTP/1.1
233	21.957886	8.18.65.32	192.168.1.10	HTTP/X.	817	HTTP/1.1 200 OK
235	21.983023	192.168.1.10	66.235.132.121	HTTP	576	GET /b/ss/applesuperglobal/1/G.6--NS?pcr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.
237	22.052601	66.235.132.121	192.168.1.10	HTTP	643	HTTP/1.1 200 OK (GIF89a)
276	25.405800	192.168.1.10	8.18.65.32	HTTP	388	GET /WebObjects/MZSearch.wa/wa/incrementalSearch?media=movie&q=hack HTTP/1.1
279	25.478572	8.18.65.32	192.168.1.10	HTTP/X.	784	HTTP/1.1 200 OK
281	25.505888	192.168.1.10	66.235.132.121	HTTP	579	GET /b/ss/applesuperglobal/1/G.6--NS?pcr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.
282	25.569916	66.235.132.121	192.168.1.10	HTTP	643	HTTP/1.1 200 OK (GIF89a)
871	54.915772	192.168.1.10	8.18.65.32	HTTP	385	GET /WebObjects/MZSearch.wa/wa/incrementalSearch?media=movie&q=s HTTP/1.1
883	55.046379	8.18.65.32	192.168.1.10	HTTP/X.	1109	HTTP/1.1 200 OK
933	55.837838	192.168.1.10	66.235.132.121	HTTP	570	GET /b/ss/applesuperglobal/1/G.6--NS?pcr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.
947	55.895278	66.235.132.121	192.168.1.10	HTTP	642	HTTP/1.1 200 OK (GIF89a)
1003	57.322835	192.168.1.10	8.18.65.32	HTTP	386	GET /WebObjects/MZSearch.wa/wa/incrementalSearch?media=movie&q=sne HTTP/1.1
1007	57.393516	8.18.65.32	192.168.1.10	HTTP/X.	1166	HTTP/1.1 200 OK
1009	57.507531	192.168.1.10	66.235.132.121	HTTP	573	GET /b/ss/applesuperglobal/1/G.6--NS?pcr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.
1011	57.556593	66.235.132.121	192.168.1.10	HTTP	642	HTTP/1.1 200 OK (GIF89a)
1119	60.633662	192.168.1.10	8.18.65.32	HTTP	387	GET /WebObjects/MZSearch.wa/wa/incrementalSearch?media=movie&q=sne HTTP/1.1
1122	60.809323	8.18.65.32	192.168.1.10	HTTP/X.	551	HTTP/1.1 200 OK
1124	60.826026	192.168.1.10	66.235.132.121	HTTP	576	GET /b/ss/applesuperglobal/1/G.6--NS?pcr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.
1125	60.878926	66.235.132.121	192.168.1.10	HTTP	643	HTTP/1.1 200 OK (GIF89a)

Figure 2 is a screenshot of the Follow HTTP Stream from frame 307. It shows the user-agent AppleTV/2.4 (1) connecting to the host ax.itunes.apple.com (2), and looking at the page for the Hackers movie by Iain Softley (3).

Figure 2 - Follow HTTP to Hackers

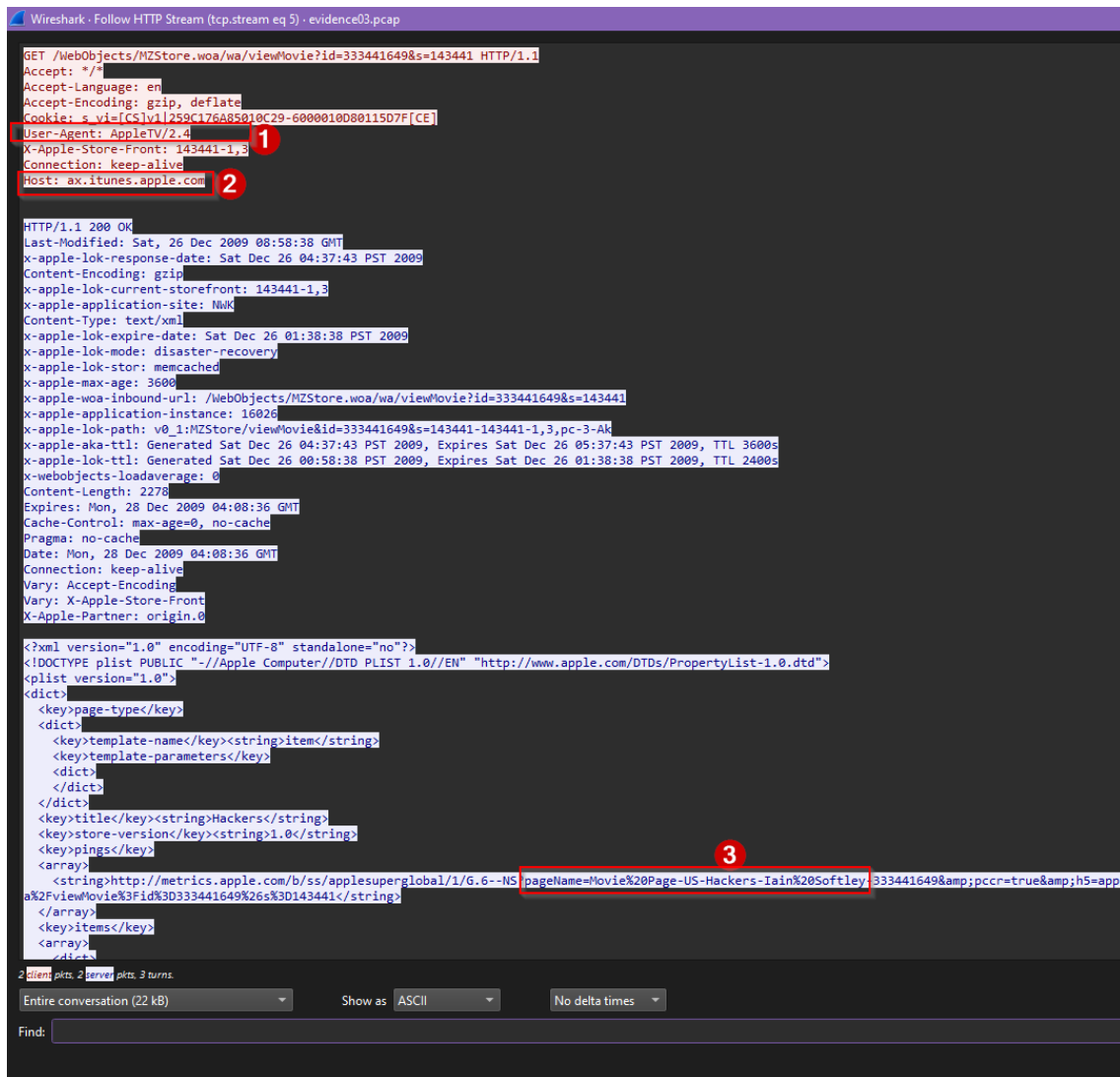


Figure 3 shows a screenshot of the word “sneak” being typed one letter at a time, over the span of ~29 seconds. Frame 1146 includes a typo of “sneb” (1), which is corrected in the next search.

Figure 3 - Sneak search

282	25.569916	66.235.132.121	192.168.1.10	HTTP	643	HTTP/1.1 200 OK (GIF89a)
871	54.915772	192.168.1.10	8.18.65.27	HTTP	385	GET /WebObjects/MZSearch.woa/wa/incrementalSearch?media=movie&q=s HTTP/1.1
883	55.046379	8.18.65.27	192.168.1.10	HTTP/X.	1109	HTTP/1.1 200 OK
933	55.837838	192.168.1.10	66.235.132.121	HTTP	570	GET /b/ss/applesuperglobal/1/G.6--NS?pccr=true&ch=Movies-Search&g=http%3A%2F%2Fax.sear
947	55.895278	66.235.132.121	192.168.1.10	HTTP	642	HTTP/1.1 200 OK (GIF89a)
1003	57.322835	192.168.1.10	8.18.65.27	HTTP	386	GET /WebObjects/MZSearch.woa/wa/incrementalSearch?media=movie&q=sn HTTP/1.1
1007	57.393516	8.18.65.27	192.168.1.10	HTTP/X.	1166	HTTP/1.1 200 OK
1009	57.507531	192.168.1.10	66.235.132.121	HTTP	573	GET /b/ss/applesuperglobal/1/G.6--NS?pccr=true&ch=Movies-Search&g=http%3A%2F%2Fax.sear
1011	57.556593	66.235.132.121	192.168.1.10	HTTP	642	HTTP/1.1 200 OK (GIF89a)
1119	60.633662	192.168.1.10	8.18.65.27	HTTP	387	GET /WebObjects/MZSearch.woa/wa/incrementalSearch?media=movie&q=sne HTTP/1.1
1122	60.809323	8.18.65.27	192.168.1.10	HTTP/X.	551	HTTP/1.1 200 OK
1124	60.826026	192.168.1.10	66.235.132.121	HTTP	576	GET /b/ss/applesuperglobal/1/G.6--NS?pccr=true&ch=Movies-Search&g=http%3A%2F%2Fax.sear
1125	60.878926	66.235.132.121	192.168.1.10	HTTP	643	HTTP/1.1 200 OK (GIF89a)
1146	71.662030	192.168.1.10	8.18.65.27	HTTP	388	GET /WebObjects/MZSearch.woa/wa/incrementalSearch?media=movie&q=sneb HTTP/1.1
1148	71.746718	8.18.65.27	192.168.1.10	HTTP/X.	1112	HTTP/1.1 200 OK
1153	71.812102	192.168.1.10	66.235.132.121	HTTP	579	GET /b/ss/applesuperglobal/1/G.6--NS?pccr=true&ch=Movies-Search&g=http%3A%2F%2Fax.sear
1154	71.871002	66.235.132.121	192.168.1.10	HTTP	643	HTTP/1.1 200 OK (GIF89a)
1157	79.755630	192.168.1.10	8.18.65.27	HTTP	388	GET /WebObjects/MZSearch.woa/wa/incrementalSearch?media=movie&q=snea HTTP/1.1
1160	79.850422	8.18.65.27	192.168.1.10	HTTP/X.	494	HTTP/1.1 200 OK
1165	79.950315	192.168.1.10	66.235.132.121	HTTP	579	GET /b/ss/applesuperglobal/1/G.6--NS?pccr=true&ch=Movies-Search&g=http%3A%2F%2Fax.sear
1166	80.010848	66.235.132.121	192.168.1.10	HTTP	643	HTTP/1.1 200 OK (GIF89a)
1168	83.874492	192.168.1.10	8.18.65.27	HTTP	389	GET /WebObjects/MZSearch.woa/wa/incrementalSearch?media=movie&q=sneak HTTP/1.1
1171	83.978382	8.18.65.27	192.168.1.10	HTTP/X.	466	HTTP/1.1 200 OK

Figure 4 is a screenshot of the Follow HTTP Stream from frame 1181. It shows the user-agent AppleTV/2.4 (1) connecting to the host ax.itunes.apple.com (2), and looking at the page for the Sneakers movie by Phil Alden Robinson (3).

Figure 4 - Follow HTTP to Sneakers

The screenshot shows the 'Follow HTTP Stream' window in Wireshark. The request line is 'GET /WebObjects/MZStore.woa/wa/viewMovie?id=283963264&s=143441 HTTP/1.1'. The 'User-Agent' header is 'AppleTV/2.4' (marked with a red circle and '1'). The 'Host' header is 'ax.itunes.apple.com' (marked with a red circle and '2'). The response status is 'HTTP/1.1 200 OK'. The 'Content-Type' is 'text/xml'. The XML body contains a plist with a title 'Sneakers' (marked with a red circle and '3') and a URL to a metrics page. The bottom status bar shows '2 client pkts, 2 server pkts, 3 turns'.

```
GET /WebObjects/MZStore.woa/wa/viewMovie?id=283963264&s=143441 HTTP/1.1
Accept: */*
Accept-Language: en
Accept-Encoding: gzip, deflate
Cookie: s_vi=(CS)vl1259C176A85010C29-6000010D8011507F[CE]
User-Agent: AppleTV/2.4
X-Apple-Store-Front: 143441-1,3
Connection: keep-alive
Host: ax.itunes.apple.com

HTTP/1.1 200 OK
Last-Modified: Sat, 26 Dec 2009 04:51:14 GMT
x-apple-lok-response-date: Sat Dec 26 02:33:51 PST 2009
Content-Encoding: gzip
x-apple-lok-current-storefront: 143441-1,3
x-apple-application-site: CUP
Content-Type: text/xml
x-apple-lok-expire-date: Fri Dec 25 21:31:14 PST 2009
x-apple-lok-mode: disaster-recovery
x-apple-lok-stor: memcached
x-apple-max-age: 3600
x-apple-woa-inbound-url: /WebObjects/MZStore.woa/wa/viewMovie?id=283963264&s=143441
x-apple-application-instance: 3127
x-apple-lok-path: v0_1:MZStore/viewMovie&id=283963264&s=143441-143441-1,3,pc-3-Ak
x-apple-aka-ttl: Generated Sat Dec 26 02:33:51 PST 2009, Expires Sat Dec 26 03:33:51 PST 2009, TTL 3600s
x-apple-lok-ttl: Generated Fri Dec 25 20:51:14 PST 2009, Expires Fri Dec 25 21:31:14 PST 2009, TTL 2400s
x-webobjects-loadaverage: 0
Content-Length: 2586
Vary: Accept-Encoding
Vary: X-Apple-Store-Front
Expires: Mon, 28 Dec 2009 04:09:29 GMT
Cache-Control: max-age=0, no-cache
Pragma: no-cache
Date: Mon, 28 Dec 2009 04:09:29 GMT
Connection: keep-alive
Vary: X-Apple-Store-Front
X-Apple-Partner: origin.0

<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<!DOCTYPE plist PUBLIC "-//Apple Computer//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
  <key>page-type</key>
  <dict>
    <key>template-name</key><string>item</string>
    <key>template-parameters</key>
    <dict>
      </dict>
    </dict>
  </dict>
  <key>title</key><string>Sneakers</string>
  <key>store-version</key><string>1.0</string>
  <key>pings</key>
  <array>
    <string>http://metrics.apple.com/b/ss/applesuperglobal/1/G.6--NS:pageName=Movie%20Page-US-Sneakers-Phil%20Alden%20Robinson 283963264&
    re.woa%2Fwa%2FviewMovie%3Fid%3D283963264%26s%3D143441</string>
  </array>
  <key>items</key>
  <array>
  </array>
</dict>
</plist>
```

Figure 5 is a screenshot of the phrase “iknowyourewatchingme” (1) being typed from frame 1472 to 1766, over the span of ~60 seconds. This suggests that the user suspects that the network traffic being monitored.

Figure 5 - Iknowyourewatchingme search

http.request.uri contains "incrementalSearch"						
No.	Time	Source	Destination	Protocol	Length	Info
1727	145.965709	8.18.65.89	192.168.1.10	HTTP/X_	1148	HTTP/1.1 200 OK
1729	145.978834	192.168.1.10	66.235.132.121	HTTP	609	GET /b/ss/applesuperglobal/1/G.6--NS?pcrr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.apple.com/1.1/ HTTP/1.1 200 OK (GIF89a)
1730	146.028253	66.235.132.121	192.168.1.10	HTTP	642	HTTP/1.1 200 OK (GIF89a)
1732	147.695922	192.168.1.10	8.18.65.89	HTTP	399	GET /WebObjects/MZSearch.wa/wa/incrementalSearch?media=movie&q=iknowyourewatch HTTP/1.1
1734	147.754741	8.18.65.89	192.168.1.10	HTTP/X_	1150	HTTP/1.1 200 OK
1736	147.763282	192.168.1.10	66.235.132.121	HTTP	612	GET /b/ss/applesuperglobal/1/G.6--NS?pcrr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.apple.com/1.1/ HTTP/1.1 200 OK (GIF89a)
1737	147.812336	66.235.132.121	192.168.1.10	HTTP	642	HTTP/1.1 200 OK (GIF89a)
1739	149.188337	192.168.1.10	8.18.65.89	HTTP	400	GET /WebObjects/MZSearch.wa/wa/incrementalSearch?media=movie&q=iknowyourewatchi HTTP/1.1
1740	149.266784	8.18.65.89	192.168.1.10	HTTP/X_	1150	HTTP/1.1 200 OK
1742	149.275723	192.168.1.10	66.235.132.121	HTTP	615	GET /b/ss/applesuperglobal/1/G.6--NS?pcrr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.apple.com/1.1/ HTTP/1.1 200 OK (GIF89a)
1743	149.323565	66.235.132.121	192.168.1.10	HTTP	641	HTTP/1.1 200 OK (GIF89a)
1745	150.824096	192.168.1.10	8.18.65.89	HTTP	401	GET /WebObjects/MZSearch.wa/wa/incrementalSearch?media=movie&q=iknowyourewatchin HTTP/1.1
1747	151.013977	8.18.65.89	192.168.1.10	HTTP/X_	1153	HTTP/1.1 200 OK
1749	151.023692	192.168.1.10	66.235.132.121	HTTP	618	GET /b/ss/applesuperglobal/1/G.6--NS?pcrr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.apple.com/1.1/ HTTP/1.1 200 OK (GIF89a)
1750	151.072481	66.235.132.121	192.168.1.10	HTTP	643	HTTP/1.1 200 OK (GIF89a)
1752	152.706244	192.168.1.10	8.18.65.89	HTTP	402	GET /WebObjects/MZSearch.wa/wa/incrementalSearch?media=movie&q=iknowyourewatching HTTP/1.1
1754	152.945869	8.18.65.89	192.168.1.10	HTTP/X_	1152	HTTP/1.1 200 OK
1756	152.957753	192.168.1.10	66.235.132.121	HTTP	621	GET /b/ss/applesuperglobal/1/G.6--NS?pcrr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.apple.com/1.1/ HTTP/1.1 200 OK (GIF89a)
1757	153.011400	66.235.132.121	192.168.1.10	HTTP	642	HTTP/1.1 200 OK (GIF89a)
1759	154.250840	192.168.1.10	8.18.65.89	HTTP	403	GET /WebObjects/MZSearch.wa/wa/incrementalSearch?media=movie&q=iknowyourewatchingm HTTP/1.1
1761	154.307985	8.18.65.89	192.168.1.10	HTTP/X_	1155	HTTP/1.1 200 OK
1763	154.315833	192.168.1.10	66.235.132.121	HTTP	624	GET /b/ss/applesuperglobal/1/G.6--NS?pcrr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.apple.com/1.1/ HTTP/1.1 200 OK (GIF89a)
1764	154.368147	66.235.132.121	192.168.1.10	HTTP	642	HTTP/1.1 200 OK (GIF89a)
1766	158.251617	192.168.1.10	8.18.65.89	HTTP	404	GET /WebObjects/MZSearch.wa/wa/incrementalSearch?media=movie&q=iknowyourewatchingme HTTP/1.1
1769	158.362253	8.18.65.89	192.168.1.10	HTTP/X_	170	HTTP/1.1 200 OK
1771	158.371395	192.168.1.10	66.235.132.121	HTTP	627	GET /b/ss/applesuperglobal/1/G.6--NS?pcrr=true&ch=Movies-Search&g=http%3A%2F%2Fsearch.itunes.apple.com/1.1/ HTTP/1.1 200 OK (GIF89a)
1772	158.420464	66.235.132.121	192.168.1.10	HTTP	643	HTTP/1.1 200 OK (GIF89a)

Total time taken: 7 hours 15 minutes.